

FINAL EXAMINATION OF SHANGHAI JIAO TONG UNIVERSITY (A)
(2019 – 2020 First Semester)

Class: _____ Student No.: _____ Name: _____
Course: Discrete Mathematics (MA208) Score: _____

Problem 1. (Logic, 14') **Prove** that $(\forall x P(x)) \vee (\exists x Q(x))$ is logically equivalent to $\forall x \exists y (P(x) \vee Q(y))$, where all the quantifiers and variables have the same **nonempty** domain.

Problem 2. (Set Theory, 14') **Prove** that for any sets A and B , $\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$, where $\mathcal{P}(\cdot)$ is the power-set operation.

I pledge that
I will obey
examination
disciplines.
Signature:

Number	1	2	3	4	5	6	7	8
Score								
Signature								

Problem 3. (Equivalence Relations, 14') Let R_1, R_2 be equivalence relations on a nonempty set A . Let P_i (for $i = 1, 2$) be the set of equivalence classes of R_i . Define $P := \{B_1 \cap B_2 \mid B_1 \in P_1, B_2 \in P_2, B_1 \cap B_2 \neq \emptyset\}$.

- (a) **Prove** that P is a partition on A ; (6')
 (b) **Prove** that the set of equivalence classes of $R_1 \cap R_2$ is equal to P . (8')

Note: You **don't** need to prove that $R_1 \cap R_2$ is an equivalence relation on A .

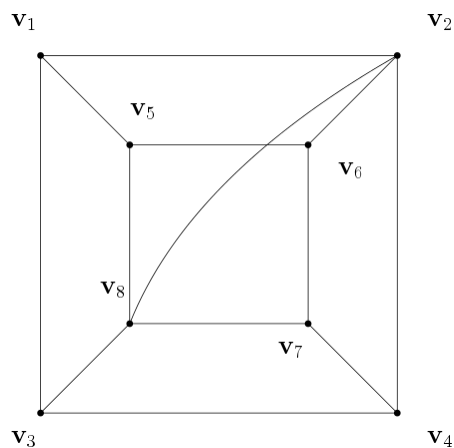
Problem 4. (Bi-transitivity, 11') Let A be any nonempty set. A binary relation R on A is called **bi-transitive** if for all $a, b, c, d \in A$, if $b \neq c$ and $(a, b), (a, c), (b, d), (c, d) \in R$, then $(a, d) \in R$. Given a binary relation R on A , we define its **bi-transitive closure** $\text{btc}(R)$ as the smallest bi-transitive relation R' on A such that $R \subseteq R'$, **if it exists**. In other words, $\text{btc}(R)$ (**if exists**) is the unique relation R' such that (i) $R \subseteq R'$, (ii) R' is bi-transitive and (iii) $R' \subseteq R''$ for all bi-transitive relations R'' on A .

- (a) Consider that $A = \mathbb{N}$ and $R = \{(n, m) \in \mathbb{N} \times \mathbb{N} \mid m = n + 1\}$. **Find out explicitly** the bi-transitive closure $\text{btc}(R)$. You **don't** need to give any proof here. (4')
 (b) **Determine** whether $\text{btc}(R)$ exists for any binary relation R or not. Please either give a **proof** or **disprove with a counterexample**. (7')

FINAL EXAMINATION OF SHANGHAI JIAO TONG UNIVERSITY (A)
(2019 – 2020 First Semester)

Class: _____ Student No.: _____ Name: _____
Course: Discrete Mathematics (MA208) Score: _____

Problem 5. (Planar Graphs, 14') Prove through **Kuratowski's theorem** that the simple graph with vertices $v_1, \dots, v_8$ below is non-planar.



Problem 6. (n -Cubes, 14') Recall that for each natural number $n \geq 1$, the n -cube Q_n is defined as a simple graph in the way that (i) its vertex set consists of all bit-strings (i.e., 0-1 strings) of length n , and (ii) there is an edge between two bit-strings iff the two bit-strings differ in exactly one bit-position.

- (a) **Prove** that Q_n is always bipartite for all $n \geq 1$. (7')
(b) **Determine** the edge connectivity $\lambda(Q_n)$ for each $n \geq 1$ and **prove** your answer. (7')

Problem 7. (Huffman Coding, 11') Consider 6 letters A, B, C, D, E, F with given (nonzero) frequencies. Suppose that (*) under one binary tree of Huffman coding, the coding lengths of A, B, C, D, E, F are 2, 2, 3, 3, 3, 3, respectively. Furthermore, (**) under another different binary tree of Huffman coding, the coding lengths of A, B, C, D, E, F are 1, 2, 4, 4, 4, 4, respectively. Please provide a solution to the frequencies of A, B, C, D, E, F and two corresponding (different) binary trees of Huffman coding that fulfill (*) and (**).

Note: If you find only one corresponding binary tree, you get 4 **points**.

Problem 8. (The Inclusion-Exclusion Principle, 8') Calculate through the **inclusion-exclusion principle** the number of complete matchings from $\{u_1, \dots, u_4\}$ to $\{v_1, \dots, v_4\}$ in the bipartite graph below. Write out the detailed application of the inclusion-exclusion principle.

